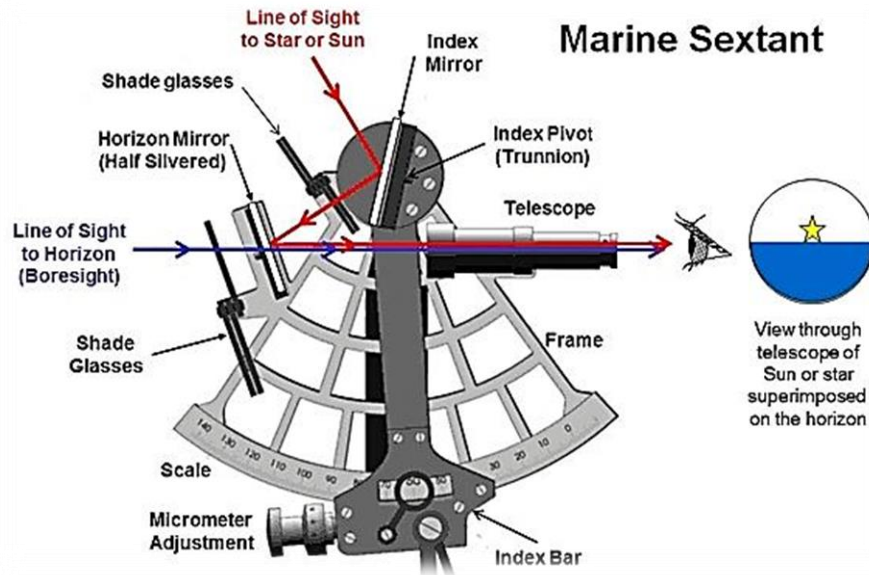


A short document on Celestial Navigation from a “flat Earther’s” perspective.

2023-2024 Version 0.2 (edit.7)

Celestial Navigation is based on acquiring elevation angles to Celestials with respect to the observer’s horizontal or horizon. The celestials are assumed to be at infinity, yet conceptualised on to the inside of a Celestial Sphere centred on the observer, based on their perspective of the sky at their geographic position & time.



Sextants are an angle measuring tool used in Celestial Navigation:

- The navigator will first determine their elevation above sea level as it’s dependent on the vessel due to how high the particular navigator’s deck is from sea level, changes are dependent on several factors, but this will serve to later level the observer’s eye with the horizon/sea level when taking an angular sextant reading.
- The navigator will be correcting for their height of eye above sea level, which is also called “dip correction”.
- If the navigator does not correct for height of eye, then they won’t achieve 90 degrees for themselves & therefore not for the GP of the Celestial either.
- The dip correction must be done unless they’re using a bubble sextant (not usually done at sea) or an artificial horizon (not usually done at sea).

» <https://astrolabesailing.com/2016/10/10/celestial-navigation-sextant-angle-corrections/>

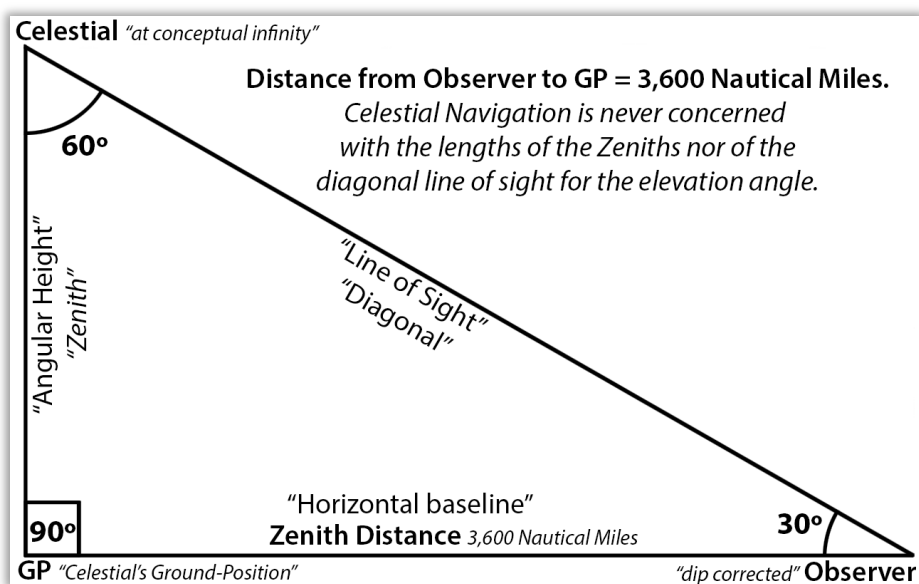
Dip – D or dip – Height of Eye Correction

If you are taking your sextant readings from the bridge of a big ship then you are going to get a different sextant reading to if you were on a small yacht. Therefore we need to make a correction to what your sextant says to get your reading converted to a sea level reading. This is called Dip or an adjustment for your height of eye.

Some Information on the Sextant and elevation angle measurements:

- With a sextant, elevation angle measurements to the Sun, Polaris or any other Celestial first require a horizontal baseline, established by sighting the horizon in the “horizon mirror”, sighting the celestial in the “index mirror” & bringing it down to the horizon by moving the “index arm”, then rocking the sextant side to side making the celestial “kiss” the horizon, and adjusting accordingly, then taking the angular reading or shouting “mark” to a companion marking the time for the sighting.

- A line of sight to the Celestial from the Observer's Position, will provide an apparent elevation of the Celestial above the Observer's horizon, giving them a measurable angle (for example 30 degrees), between the horizontal baseline established and the Celestial targeted.
 - Knowing there's another 90 degree position along the horizontal baseline, this position is labelled the GP of the Celestial or the "Ground or Geographic Position", which is the point at which the Celestial would be directly overhead 90 degrees to the horizontal baseline.
 - The Observer, the horizon & the Celestial's GP are located on the surface of Earth we're navigating, which is the prime horizontal baseline for all horizontals and level & therefore perpendicular verticals too.
- You'll now want to calculate, 90 degrees minus 30 degrees (measured elevation angle) equals the remaining 3rd angle, completing the right angle triangle, which will always be (or be parallel to) the surface of Earth no matter the length of the "Adjacent" (horizontal baseline).



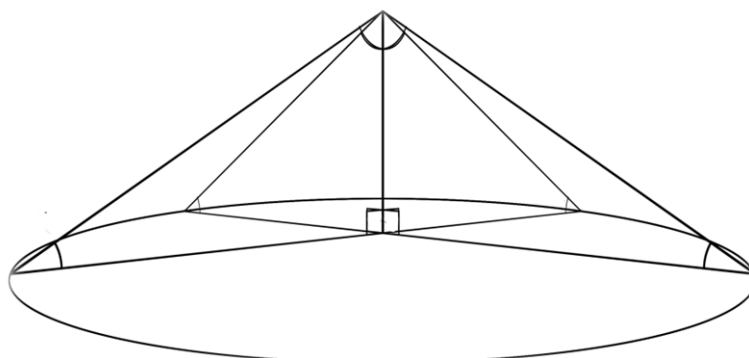
Elevation Angle and the length of the Horizontal Adjacent:

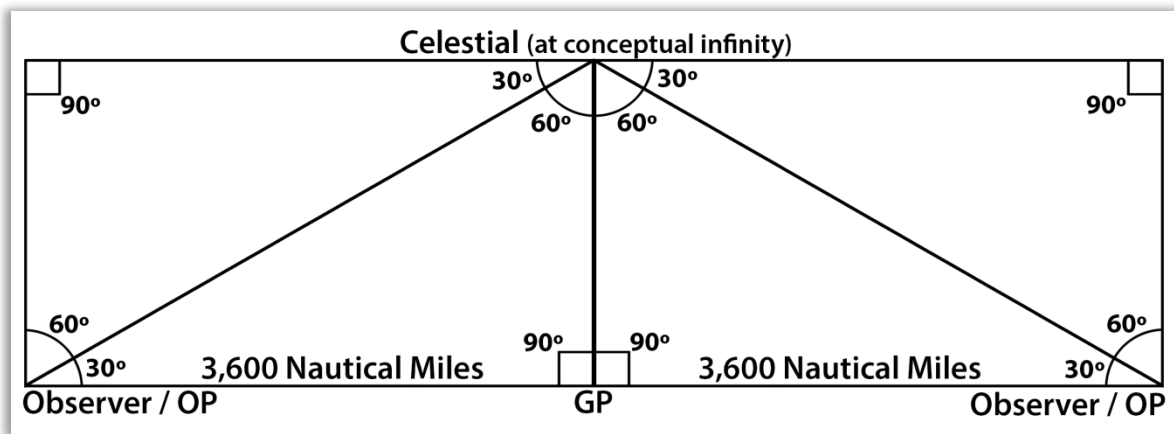
- 1 degree of arc (in the sky) is equal to 60 nautical miles on Earth's surface.
- If measured angle is 30 degrees. $90 - 30 = 60$
- 60 degrees x 60 Nautical Miles (minutes of degree) = 3,600 Nautical Miles.
- 3,600NM from Observer to Celestial's GP over a flat horizontal plane.

Circles of equal altitude, they're circles of position, or if you're anywhere on the circle, you'll have the same angle measured as someone else on the other side of the circle, or at any other point on that circle, meaning both or all observers on such circle are equidistant to the Celestial's GP.

All observers on the circle will have the same distance to GP, thus they each have the value for the radius of the circle of equal altitude. (Example below)

- 4 Observers sighting a celestial on the same C.o.E.A. (Circle of Equal Altitude).

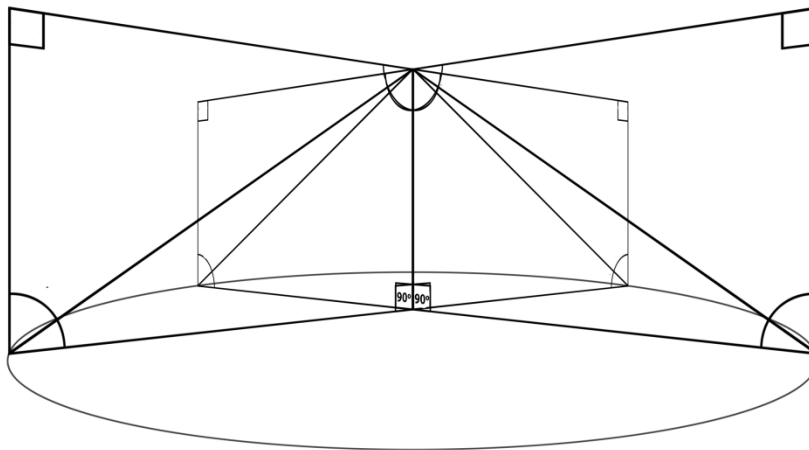




- $3,600\text{NM} \times 2 = 7,200\text{NM}$.

- A 30 degree elevation angle measurement will result in a circle of equal altitude with a diameter of 7,200 Nautical Miles, which converts to 8,285.612 Miles.

- Did you know “the ball Earth” is claimed to be 8,000 Miles wide? *Oops!* We just made a circle wider than the presupposed ball itself is claimed to be. *Do we bend that circle? (Sarcasm).*



To detail the Circle of Equal Altitude and the method mentioned above;

https://www.starpath.com/cgi-bin/web_card/courses/glossary.pl?show_def=26&cat=Celestial_Navigation

Circle of Equal Altitude

Any circle of points on the earth's surface that is equal distance from the geographical position (GP) of a celestial body at a precise moment of time. All observers located anywhere on this circle would measure the same sextant height of the body at that moment. The center of the circle is the GP of the body, the radius of the circle is the zenith distance (z). The Ho they would observe is $H_o = 90^\circ - z$. When taking the sights, the observers would all be looking toward the body in the center of the circle, which means some would be looking north, others, south or west, etc.. If the body were 60° high, the zenith distance would be 30° , which means the distance to the GP would be $30^\circ \times (60 \text{ miles}/1^\circ)$, or 1,800 miles. The LOPs used in celestial navigation are short segments of the circumference of this circle, which we approximate as straight lines.

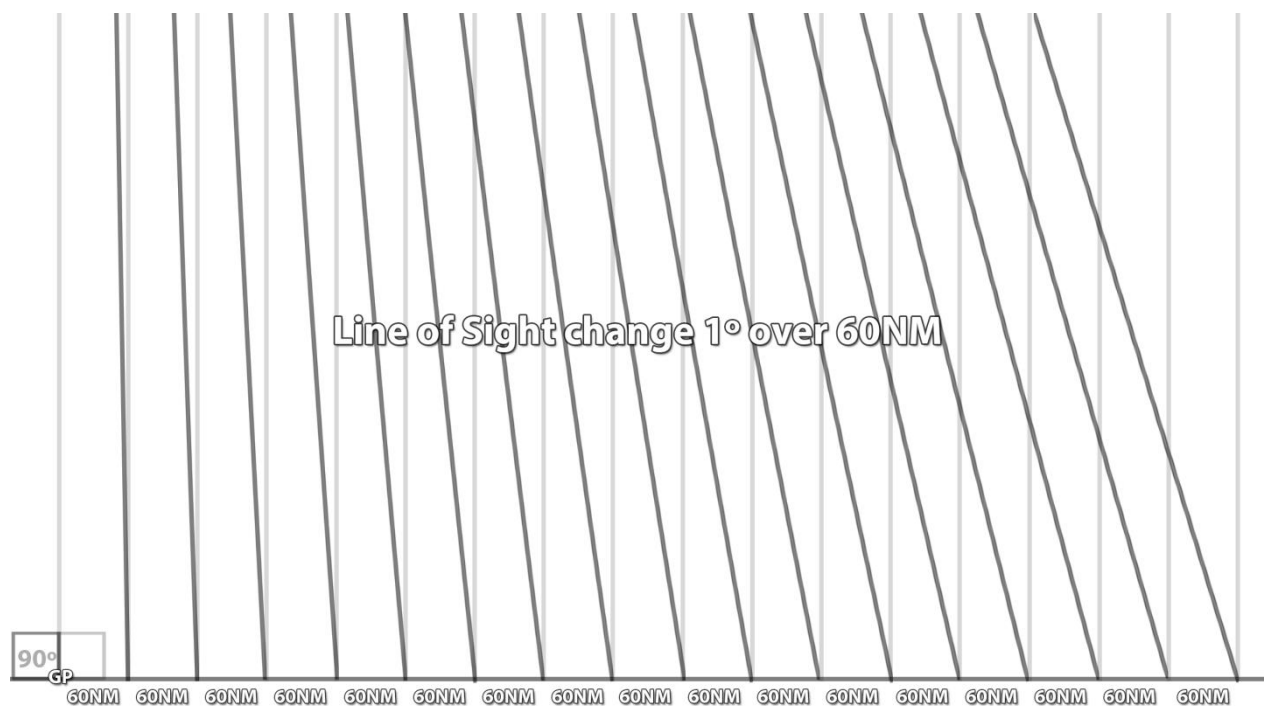
- To my current knowledge, circles of equal altitude are not accurately drawn on maps as the maps are distorted from having been projected on to a sphere (or other shape) before projecting them as flat maps & therefore globe defenders take to drawing circles on their desktop globes to match with coordinates intended to be used as a 2D grid system on flat maps. (Hence the need for the haversine formula in the Almanacs)

- A navigator will draw a LoP or a Line of Position on a plotting sheet, which is drawn as a straight line, which is representing a small section of the circumference of said “circle of equal altitude”.

- In order to perform a “fix” a navigator must sight two or three celestials and mark out their circles of equal altitude & the intersection of those circles will provide them with their rough position.

- Depending on accuracy of measurement and adequate correction performed, a navigator can expect to be within one degree or 60 Nautical Miles from their “Estimated Position” to their “Calculated Position”.

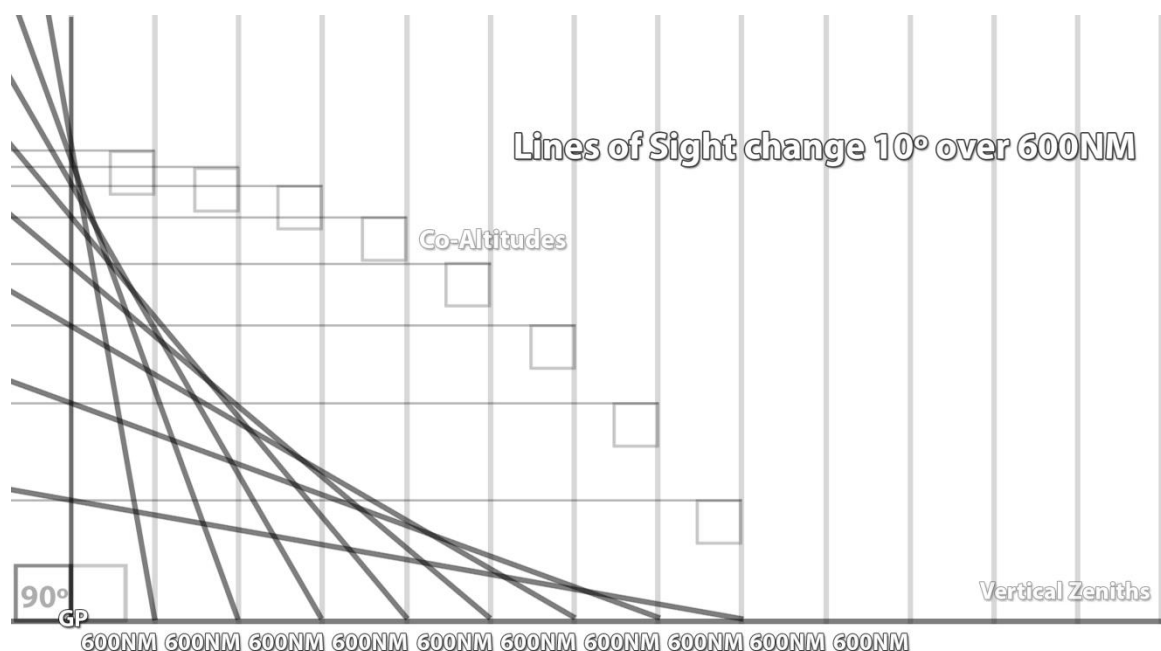
Please note, I depict the Celestials as being “at conceptual infinity” as described in the Celestial Sphere model, though I get asked to make the Hypotenuse almost parallel with the Celestial’s Zenith as that’s how it’s depicted for a globe with a 93million mile sun. Note the contradiction, they ask to elevate beyond infinity to reach a lower number of 93million miles. Simply because they want to deny it being a triangle, even though it still is one in their paradigm too, just cutting through a hump.



As you can see, the first degree away from the GP (60 NM) regardless of how far away the Celestial truly is, the diagonal “Hypotenuse” will always end up intersecting the Celestial’s Zenith & therefore forming a right-triangle. (Assuming the point of intersection is the point at which the Celestial is apparent to the observer)

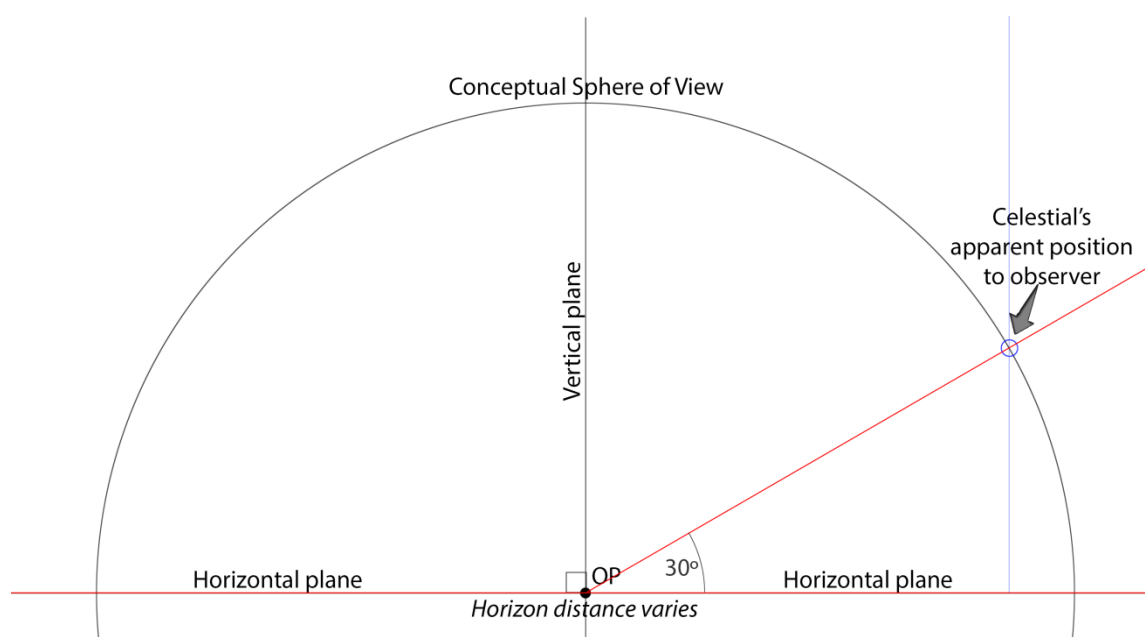
- The thing about lines or rays & how angles are measured, one could extend all lines and still have the same measurable angle, the same is not true for arcs or curves.

Here you find a “zoomed out” version of the above, changing from 1 degree per 60 Nautical Miles, to 10 degrees per 600 Nautical Miles, with added Co-Altitudes detailing the points of intersection with the Celestial’s Zenith & therefore detailing an apparent angular height for the Celestial.

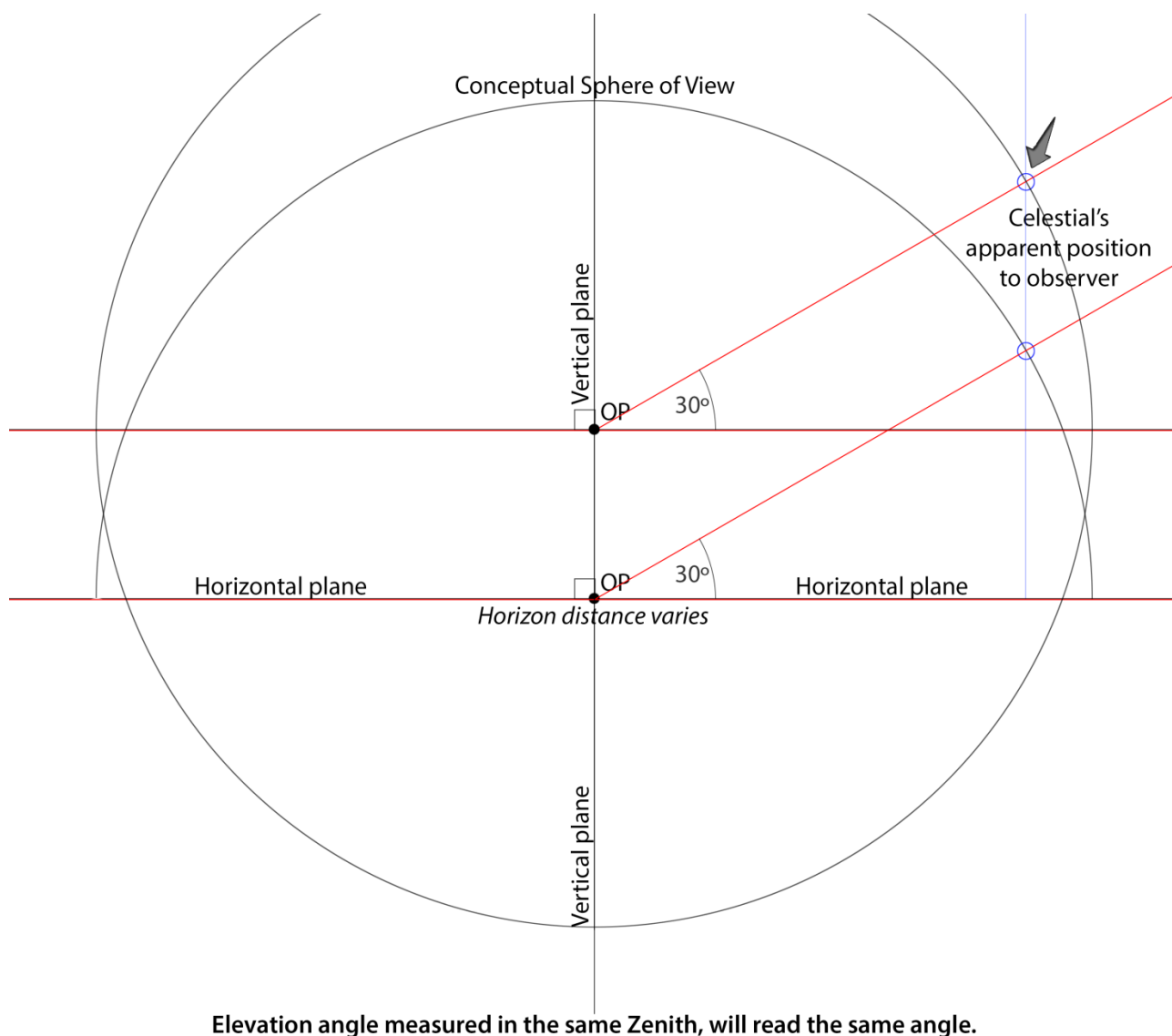


Note that as the Observer approaches the Celestial GP, the elevation angle increases, and the Celestial will appear to drop in elevation above the horizon as the Observer or Celestial move away.

The observer’s perspective in relation to the horizon and the sky above can be interpreted as a personal Celestial Sphere surrounding them, or more accurately a half sphere for the sky & thus I’m representing the apparent elevation of a celestial on this apparent arc over the observer, which is acting as a protractor. 0 degrees for horizon, 90 degrees for overhead & back to 0 degrees (180°) for the horizon in the opposite direction.



When two or more observers share the same zenith, regardless of altitude between them, they'll all measure the same elevation angle value when measuring to the same celestial at the same time.



Considering the above; an aeroplane at cruising altitude will measure the same elevation angle to the same celestial at the same time as someone directly below them on the surface of Earth, as they share a common Zenith & thus common ground position on the horizontal reference of Earth's surface.

This being the case; we can assert that we're not moving closer to the celestial, the higher up an observer goes. Which is why Celestial Navigation methodology only provides us horizontal distances & not vertical nor diagonal distances.

Therefore I can also assert that this further demolishes the notion of a ball Earth, as apologists claim diverging zeniths and thus they're not vertical & that implies much greater distances between Zenith A to Zenith B the higher in altitude the aircraft go; which is not experienced.

Always parallel horizontals & parallel verticals perpendicular to the horizontal surface of Earth we dwell atop.

A brief touch on the Geometry being used in Celestial Navigation:

- In planar Euclidean Geometry, triangles have 3 straight lines meeting at 3 vertices giving you 3 measurable angles and thus always adding up to 180 degrees.
- Non-Euclidean (doesn't actually exist) "spherical triangles", claimed to be using 3 arcs meeting at 3 supposed vertices which won't give you measurable angles as we don't measure angles from curves, thus never actually adding up to more than 180 degrees.

The definition of "angle" from Wikipedia:

- *"In Euclidean geometry, an angle is the figure formed by two rays, called the sides of the angle, sharing a common endpoint, called the vertex of the angle. Angles formed by two rays are also known as plane angles as they lie in the plane that contains the rays. Angles are also formed by intersecting two planes; these are called dihedral angles."*

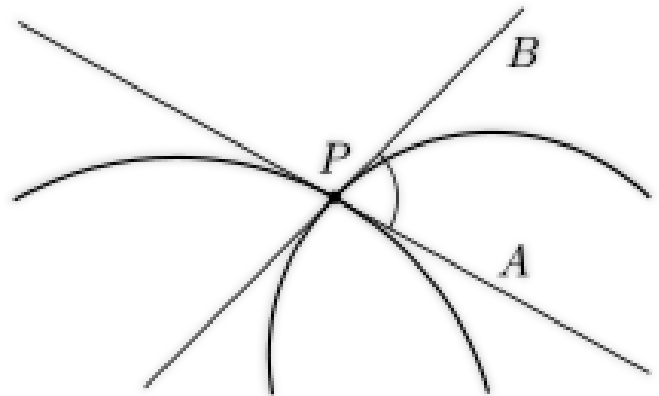
- Which is good and supporting my position, the following statement from the wiki is also correct, however I feel I should give it a little attention and clear up any misconceptions the reader may have when reading the following quote.

- *"Two intersecting curves may also define an angle, which is the angle of the rays lying tangent to the respective curves at their point of intersection."*

- This is not "non-Euclidean geometry", this is not simply two arcs or curves giving an angle, this is two straight-lines tangential to two arcs (one each), and the straight-lines intersecting where the arcs intersect, the straight-lines then give the angles you supposedly could after ascribe to the arcs themselves.

I assert, angles are not measured with arcs.

And; Complementary angles add to 90 degrees &
Supplementary angles add to 180 degrees.



My message, intentions and disclaimers:

- I'm documenting my findings on various topics within the "flat Earth" subject, with a series of downloadable pdf-files to then share in digital and/or physical formats, maybe even eventually compile a small book out of it all.
- With updates, later editions of these documents should be more detailed & comprehensive for the average reader.
- I number each document every time I update it, I mark them with an alias name, document title, version and year so that the reader has the ability to identify & download the latest one.

A message to my peers:

- I wish to inspire others to do the same, but I feel I must lead with example, no one needs to hear *"please write everything you just said down & make charts and diagrams so I can better understand your position"*, but I found myself drawing on images and pinning copies of my responses in my notes to save time detailing points I kept finding my "debate opponents" not comprehending or re-asking ad-nauseam.
- Whilst making a simple diagram, I thought, *why isn't "everyone" trying to concisely detail flat Earth sub-topics and points of contention in document format?* I appreciate that I am not privy to all documents ever created by all "flat Earthers" ever, and I'm not claiming to be the pioneer of making a pdf-document either, but for my intended purposes, "opponent's requirements" and my wishes to inform people, I don't see a reason why "we" shouldn't be creating these kinds of documents and making them available for free to the average person, share them with your peers, print them out and share them with family or friends. No family member of mine is interested in multi-hour long podcasts (regardless of the topic).
- I'd like to suggest documenting the topics you're most interested and/or proficient in, share them with your peers, update & refine to make them as concise but informative as possible & then share them with your peers again for further constructive criticism, to ultimately collectively organise a mass of documents detailing every possible sub-topic of "flat Earth". This way we "pave the paths" for people that are new to the topic of "flat Earth", rather than expecting them to sift through hundreds of thousands of hours of content to maybe only receive broken messages, half understandings and/or fragments of arguments.

One should always do their own research to validate the content for themselves, this way bad documents get exposed &/or corrected. (peer-review)

My personal Links:

- The Hub (Discord Server), you can find & download my PDFs in here [#downloads](#) (free to download & free to use) (includes documents on: "Relative Density", "Scientific Method", "Gas pressure prerequisite container" ...)
- You can contact me through Discord, via [The Hub](#) server for any corrections or additions etc..
- Github archive coming soon! (this is so ANYONE can find these documents, with zero requirement of a user account, as you can only see INSIDE of my Discord server, if you have a Discord account... which is NOT ideal)

Thank you for reading

@TM_theviking